

Update to the management of recurrent urinary tract infections in women aged 16 years and older



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Abstract

Recurrent urinary tract infections (rUTIs) are a burdensome condition affecting approximately 6% or 1.7 million women in the UK. UTIs are also a common reason for antibiotic use, with UK data demonstrating that they are the second most common reason for antibiotic prescribing after respiratory tract infections. UTIs also result in significant healthcare costs, with hospital admissions alone estimated to have cost the National Health Service in England over £600 million from 2023 to 2024. This review provides an up-to-date overview of the management of rUTIs, focusing on the updated 2024 National Institute for Health and Care Excellence guidance, which recommends a stepwise approach starting with behavioural measures, then progressing to non-antibiotic and finally antibiotic prevention. It also compares the guidance with international recommendations and outlines the current evidence on pathogenesis, diagnostic approaches and treatment options, including behavioural measures, non-antibiotic interventions (eg, vaginal oestrogen and methenamine hippurate) and antibiotic prophylaxis. Key challenges in primary care are discussed, such as diagnostic uncertainty, patient dissatisfaction and antimicrobial resistance (AMR). Emerging research in point-of-care testing, the urinary microbiome and novel therapies is also highlighted. This review aims to support safe prescribing, improve patient satisfaction and mitigate AMR by promoting antimicrobial and diagnostic stewardship in the management of rUTIs.

KEY MESSAGES

- ▶ Recurrent urinary tract infections (UTIs) are common and burdensome, particularly in women and older adults, with significant impacts on quality of life and healthcare use.
- ▶ Previous management strategies have historically relied heavily on antibiotics, despite concerns about resistance, diagnostic uncertainty and patient dissatisfaction.
- ▶ Updated National Institute for Health and Care Excellence (NICE) guidance (2024) recommends a stepwise approach to recurrent UTI (rUTI) management, starting with behavioural measures and progressing to non-antibiotic and then antibiotic options.
- ▶ Vaginal oestrogen and methenamine hippurate are highlighted in the NICE guidance as potential antibiotic-sparing options for rUTI prophylaxis.
- ▶ Tailored management and regular review of antibiotic prophylaxis can support antimicrobial stewardship and improve patient satisfaction.

Introduction

Urinary tract infections (UTIs) are among the most common bacterial infections globally, affecting an estimated 400 million people in 2019.¹ In the UK, UTIs account for a quarter of all antibiotic prescriptions, making them a key focus for antimicrobial stewardship.² The latest English surveillance programme for antimicrobial utilisation and resistance report indicates that there were over 1.8 million UTI-related hospital admissions in England from 2018–2019 to 2022–2023.³ More recent data from 2023 to 2024 indicate that there were nearly 190 000 UTI-related hospital admissions in England, costing the National Health Service (NHS) over £600 million.⁴ Although UTIs affect all ages and sexes, they

disproportionately impact older adults and women.^{5,6} Survey data from the UK and USA suggest that 37% and 60% of women, respectively, will experience at least one UTI in their lifetime.^{7,8}

Recurrence is common following a UTI, particularly in older adults and women.^{6,9} The current internationally accepted definition of recurrent UTIs (rUTIs) is two or more UTIs in 6 months, or three or more in 12 months.^{10–12} Population-based studies suggest that 15%–19% of women will develop rUTIs after a UTI, equating to approximately 6% of all adult women in the UK.^{13–15} Based on recent UK population-based data, this equates to approximately 1.7 million women being affected by rUTIs.¹⁶ This article provides a practical, evidence-based overview of rUTI management for generalist

clinicians, highlighting recent clinical guidance and emerging research.

Pathogenesis and risk factors

Mechanisms of recurrence

Currently, there are two models explaining the pathogenesis of rUTIs; however, it is not clear if either or both are responsible. The first model suggests that infections, both isolated and recurrent, arise from bacteria outside the urinary tract, such as the faecal flora, ascending up the urethra and causing infection.^{17,18} The second model suggests that, in rUTI, bacteria persist intracellularly within the urinary tract, forming reservoirs that evade immune responses and antibiotics.^{17,18} To date, evidence for this model is primarily derived from rodent studies with limited evidence in humans. The discovery of the urinary microbiome challenges the long-held belief that urine is sterile, but its role in rUTIs and how it interacts with the vaginal and gastrointestinal microbiomes is unclear.^{19–21}

Risk factors

Various risk factors have been identified for rUTIs, but these differ depending on menopausal status (box 1) and are based on few studies (table 1). Key risk factors in premenopausal women are sexual activity and UTIs in childhood or a family history of UTIs, whereas in postmenopausal women there are additional risk factors related to menopausal changes, inadequate bladder emptying and reduced physical function.^{10,22}

Diagnosis and investigation

Diagnostic approach

In the UK, the diagnosis of a UTI is primarily made using clinical assessment with additional tests, such as urine dipstick and urine culture, used only in selected cases.^{22,23} In women under 65 years, the National Institute for Health and Care Excellence (NICE) and the UK Health Security Agency (UKHSA) guidance recommend empirical treatment for suspected UTIs if two or more of the following symptoms are present: dysuria, new nocturia, or cloudy urine, and if vaginal and urethral causes are adequately ruled out.²³ This is based on evidence suggesting that in patients presenting with these symptoms, additional testing offers only minimal increases in accuracy.^{10,23} If only one symptom is present or other new symptoms such as frequency, urgency, haematuria or suprapubic tenderness are present, then urine dipstick testing may guide treatment. In women over 65 years, diagnosis is symptom-based and urine dipsticks should not be used for diagnosis due to the prevalence of asymptomatic

Table 1 Research in women on risk factors for developing recurrent urinary tract infections (rUTIs) and their effect size

Risk factor	Effect measure and size (95% CIs)
Raz <i>et al.</i> ⁶² Israel, n=212. Retrospective case–control study. Postmenopausal women (mean age=66 years)	
Urinary incontinence	OR 5.79 (2.05 to 16.42)
History of UTI before menopause	OR 4.85 (1.70 to 13.84)
Non-secretor status*	OR 2.90 (1.28 to 6.25)
Foxman <i>et al.</i> ⁶³ USA, n=224. Prospective cohort study. Women 18–39 years (no age summary available)	
Vaginal intercourse (≥1/2 weeks)	OR 1.49 (1.08 to 2.06)
Diaphragm/spermicide use	OR 1.76 (1.25 to 2.48)
Scholes <i>et al.</i> ⁶⁴ USA, n=482. Retrospective case–control study. Women aged 18–30 years (no age summary available)	
4–8 episodes of intercourse in the past month†	OR 5.8 (3.1 to 10.6)
≥9 episodes of intercourse in the past month†	OR 10.3 (5.8 to 18.3)
≤15 years at first UTI	OR 3.9 (1.9 to 8.0)
Maternal history of UTI	OR 2.3 (1.5 to 3.7)
Any spermicide use in the past year	OR 1.8 (1.1 to 2.9)
New sexual partner in the past year	OR 1.9 (1.2 to 3.2)
Nseir <i>et al.</i> ⁶⁵ Israel, n=244. Retrospective case–control study. Premenopausal women (mean age: 44 years SD 9 (cases), 40, SD 10 (controls))	
Maternal history of RUTIs	OR 1.60 (1.2 to 2.0)
No probiotic use	OR 1.90 (1.21 to 2.4)
BMI >30 kg/m ²	OR 4.00 (3.2 to 4.61)
Ackerson <i>et al.</i> ¹⁴ USA, n=374, 171. Retrospective cohort study. Women ≥18 years (median age=45 years)	
Diabetes mellitus	RR 1.07 (1.05 to 1.10)
Dementia	RR 1.20 (1.14 to 1.27)
Immunocompromised	RR 1.12 (1.07 to 1.18)
Oral contraceptive use	RR 1.12 (1.09 to 1.15)
Charlson Comorbidity Index ≥3†	RR 1.12 (1.08 to 1.17)
1–2 antibiotic prescriptions in prior year†‡	RR 1.07 (1.05 to 1.10)
≥3 antibiotic prescriptions in prior year†‡	RR 1.19 (1.15 to 1.24)
Positive urine culture	RR 1.14 (1.12 to 1.16)
Antibiotic resistance (≥3 classes)†	RR 1.08 (1.05 to 1.12)
Age ≥78 years (compared with 18–27 years)	RR 1.07 (1.03 to 1.11)
1–4 outpatient visits in prior year†	RR 1.24 (1.17 to 1.31)
5–8 outpatient visits in prior year†	RR 1.43 (1.35 to 1.52)
9–15 outpatient visits in prior year†	RR 1.54 (1.45 to 1.64)
≥16 outpatient visits in prior year†	RR 1.66 (1.56 to 1.76)
≥2 emergency department visits in the prior year†	RR 1.07 (1.04 to 1.10)

*Reduced expression of antigens on the mucosa that reduce bacterial adhesion.

†Risk factors compared with 0 (reference).

‡The study also found that those treated with nitrofurantoin, fluoroquinolones or cephalosporins were at a higher risk of developing rUTIs (compared with no exposure to these antibiotics in the previous year).

BMI, body mass index; RR, risk ratio.

Box 1 Risk factors for recurrent urinary tract infections (UTIs) according to menopausal status^{10,22}

Premenopausal women.

Sexual intercourse, spermicide use and a new sexual partner.

Maternal history of UTIs.

UTIs during childhood.

Postmenopausal women

Sexual intercourse, spermicide use and a new sexual partner.

UTIs prior to menopause.

Urogenital atrophy/atrophic vaginitis related to oestrogen deficiency.

Urinary incontinence.

Cystocele.

Inadequate bladder emptying/increased post-void volume.

Urinary catheterisation.

Reduced functional status in those in long-term care.

bacteriuria.²³ However, when a UTI is suspected, a urine sample should be obtained before starting antibiotics and sent for culture and susceptibility testing, as the risk of resistance is higher in older adults.²³ Importantly, the UKHSA diagnostic guidance is designed for acute UTIs, not rUTIs, and equivalent diagnostic criteria are currently lacking. In the context of rUTIs, NICE guidance recommends sending urine samples for bacterial culture and antibiotic susceptibility testing, but is not explicit about its role in diagnosis.¹²

The Scottish Intercollegiate Guidelines Network guideline (SIGN) differs from those described above.²⁴ It recommends that if two or more of the following symptoms are present: dysuria, frequency, urgency, visible haematuria and nocturia, then urine dipstick testing should be performed to inform the diagnosis. They do not recommend empirical treatment based on symptoms alone and recommend that if a single urinary symptom is present, then self-care advice should be provided. The SIGN guidance also recommends, similarly to the UKHSA guidance, that if new onset vaginal irritation or discharge is present (in combination with any urinary symptoms) then alternative diagnoses should be considered such as a sexually transmitted infection.^{23 24} With regard to diagnosing acute UTIs in women with rUTIs, and in contrast to the UKHSA guidance, the guidelines state that 'there is no difference in the diagnostic criteria used when managing recurrent UTI'.²⁴

A recent diagnostic accuracy study in over 1000 women presenting to UK primary care with UTI symptoms assessed the performance of both the SIGN guidance and the previous version of the UKHSA guidance.²⁵ The study found that both sets of guidance performed poorly in identifying women with a positive urine culture. Despite the limitations of urine culture for identifying uropathogens, this study suggests that guideline adherence could result in diagnostic errors in the management of uncomplicated UTIs.²⁵

Further investigation

Recommendations on further investigation in women with rUTIs are limited. NICE guidance advises considering further investigation, such as an ultrasound, where self-care and non-antibiotic forms of prevention have failed, and prior to initiation of prophylactic antibiotics.¹² A recent NHS England and UKHSA guideline provides additional advice for investigation^{26 27}:

- Blood tests to identify potential risk factors (eg, glycated haemoglobin (HbA1c), urea and electrolytes).
- Ultrasound scan and assessment of bladder residual volume.

NICE advises considering specialist referral if an underlying cause is suspected (box 2) but also when the underlying cause is unknown.¹² In some cases, it might be clear what the likely cause is (eg, oestrogen deficiency secondary to the menopause); however, in other cases, this might not be clear. This ambiguity may contribute to clinical uncertainty regarding the appropriateness of further investigations and complicate communication with patients experiencing rUTIs.

Recurrent UTIs could be a presentation of bladder cancer; therefore, the NICE guidance states that non-urgent referral should be considered in women aged 60 years and older with unexplained rUTIs or a persistent UTI.¹² Urological cancer (bladder or renal) should also be considered, and women referred on a suspected cancer pathway, if they are 45 years or older with visible haematuria that either persists or recurs following successful UTI treatment.²⁸ Additionally, women aged 45 years and older with unexplained haematuria or aged 60 years and older with unexplained non-visible haematuria with dysuria or an elevated white blood cell count on blood tests should also be referred on a suspected cancer pathway.²⁸

Box 2 Guidance about when specialist advice or referral is recommended

Refer or seek specialist advice regarding investigation.¹²

- Trans women and non-binary people with a male genitourinary system.
- Recurrent upper urinary tract infection (UTI) (ie, pyelonephritis).
- Recurrent lower UTI where the underlying cause is unknown.
- Consider a non-urgent referral in those aged 60 years and older.
- Pregnant women, trans men and non-binary people.
- People with suspected cancer.
- Previous gender reassignment surgery involving structural alteration of the urethra.

Arrange specialist referral²²

- Suspected underlying cause of recurrent UTIs including:
 - Renal/ureteric stones.
 - Interstitial cystitis.
 - Urogynaecological cancer.

Challenges in primary care

Most rUTIs are managed in primary care and present clinicians with several challenges.

Patient experience

There is increasing international evidence that women with rUTIs are frustrated with the care they receive.²⁹ Women with rUTIs often feel their condition and its impact are minimised.²⁹ Research demonstrates the significant physical, psychological and relationship impacts, yet patients perceive a lack of empathy or understanding from healthcare professionals.^{29–31} Nonetheless, evidence indicates that healthcare professionals recognise the impact of rUTIs, although this recognition may not always be effectively communicated during consultations.²⁹

Diagnostic uncertainty

Current diagnostics for UTIs are limited. Urine dipsticks have limited sensitivity and specificity, with a poor negative predictive value in symptomatic patients and urine culture is known not to detect all potential uropathogens.^{32–34} Due to the limitations of standard urine culture, enhanced urine culture exists to culture fastidious and difficult-to-culture uropathogens. However, this method is not currently included in UK guidance for UTIs, access is likely to be limited and heterogenous across the UK and it would require further discussion with microbiology laboratories. Additionally, the correlation between antibiotic susceptibility in cultured uropathogens and treatment response is imperfect.³⁵ Therefore, these tests are associated with false positives and false negatives and can drive both undertreatment and overtreatment.³³ Over-treatment drives antimicrobial resistance (AMR) and antibiotic adverse effects, including *Clostridioides difficile* infection (particularly in older adults receiving repeated antibiotic courses). Undertreatment risks progression to pyelonephritis, sepsis and death.

Currently, there are no diagnostic tests specifically for rUTIs and internationally there is varying guidance on diagnosing rUTIs and the use of urine analysis methods.^{10–12}

In view of the potential serious underlying causes for rUTIs described above, both clinicians and patients alike worry about missing these, especially when symptoms persist without clear triggers.^{29 36} Diagnostic stewardship aims to optimise the appropriate use and interpretations of diagnostic tests to guide

treatment decisions.³⁷ It is a critical component of antimicrobial stewardship because accurate diagnosis is crucial to ensure appropriate antibiotic use. Importantly, diagnostic stewardship also supports the development and adoption of improved diagnostic technologies, which can reduce uncertainty and enable faster, more precise treatment decisions. Together, these measures can help prevent unnecessary antibiotic exposure, support targeted therapy, reduce the risk of serious underlying causes being missed and improve patient safety and outcomes in rUTI management.

Antimicrobial resistance

AMR is a global health challenge, especially in the context of UTIs, and is a key priority for the UK Government.³⁸ The most common uropathogen is *E. coli*, which causes approximately 70% of all UTIs^{3 13 39} with high rates of resistance to amoxicillin (estimates from 47% to 51%) followed by trimethoprim (30%) and co-amoxiclav (23%–35%) (table 2).^{3 13 39} Bacteraemia, or bloodstream infections, is a serious infection associated with substantial mortality, and resistant bacteraemia has increased by 13.1% from 2019 to 2023 in England, equating to an average of 400 resistant cases per week.³ *Escherichia coli* is also the most commonly isolated organism in bacteraemia and UTIs are consistently reported as the most common primary focus.³ Additionally, *E. coli* accounted for 70.5% of antibiotic-resistant cases of bacteraemia in 2024.³

The impact of rUTIs is substantial, so balancing effective treatment against concerns about antibiotic resistance is challenging.^{40 41} Antimicrobial stewardship encompasses various strategies to optimise antibiotic use and minimise the unintended consequences of their use.³⁸ In rUTIs, this means UTI prevention using non-antimicrobial strategies, which includes behavioural strategies (eg, increased fluid intake) and preventatives that do not contain antibiotics (eg, vaginal oestrogen), regularly reviewing prophylaxis and reducing antibiotic exposure where appropriate. Evidence indicates that women with rUTIs often attempt to self-manage without antibiotics, both acutely and prophylactically, but are frustrated that non-antibiotic treatment options are commonly not discussed by healthcare professionals.^{29 36} Uncertainty related to the effectiveness of the different non-antibiotic options may be one of the reasons for this. However, the recently updated NICE guidance, including an emphasis on non-antibiotic prevention, might help support decision making and increase the use of non-antibiotic options.¹² This has the potential to reduce antibiotic exposure and risk of antibiotic resistance.

Management strategies

In 2024, UK guidance on the management of rUTIs by NICE was updated.¹² This section will summarise this guidance (figure 1) and refer to the primary care management of rUTIs in women, trans men or non-binary people with a female urinary system.

Self-care and behavioural measures

NICE recommends self-care as the first step in rUTI prevention. The advice includes hydration, avoiding delayed urination, wiping front to back, and avoiding douching and occlusive underwear.¹² Despite personal hygiene measures being advised, the evidence base is limited and largely based on expert guidance with some international guidelines not including them in their management guidance (apart from increased hydration).^{11 12} Increasing hydration is the most substantiated intervention, but this is limited to one clinical trial conducted in 2018.^{12 42} This US-based clinical trial demonstrated that increasing fluid intake in pre-menopausal women (n=140) drinking less than 1.5 litres per day reduced the mean number of UTIs (absolute mean difference 1.5, 95%

Table 2 Proportion of urine cultures that have grown *Escherichia coli* that have evidence of resistance for antibiotics used to treat UTIs in the community in 2024^{3 39}

		Evidence of antibiotic resistance	
		ESPAUR ³	PHW ³⁹
Antibiotic	Amoxicillin	51.4%	46.9
	Trimethoprim	30.1%	29.7%
	Co-amoxiclav	22.6%	34.7%
	Cephalexin	13.6%	9.0%*
	Ciprofloxacin	13.1%	8.6%†
	Pivmecillinam	8.3%	3.2%
	Fosfomycin	3.6%	0.9%
	Nitrofurantoin	2.2%	1.5%

Note that resistance percentages from the PHW report are based on community urine samples only whereas the ESPAUR report includes both inpatient and community urine samples.

*Includes cephalexin and/or cefradine.

†Includes ciprofloxacin and/or levofloxacin.

ESPAUR, English surveillance programme for antimicrobial utilisation and resistance; PHW, Public Health Wales; UTIs, urinary tract infections.

CI 1.2 to 1.8) and the mean number of antibiotic courses (1.7, 95% CI 1.3 to 2.1).⁴² This evidence suggests that correcting underhydration reduces rUTI recurrence, but overhydrating is also not advisable.

NICE guidance acknowledges, without recommending, that patients may consider using over-the-counter products such as D-mannose (a sugar in the form of a powder or tablet) and cranberry products (in various forms).¹² However, NICE Clinical Knowledge Summaries (CKS) do not recommend D-mannose or cranberry-based products.²² For clarity, it is important to note that although the development process of NICE CKS guidance is accredited by NICE, it is not equivalent to NICE guidance since it has not been developed using a NICE process. The evidence for these options is mixed and uncertain with Cochrane reviews showing variable results. For D-mannose, a 2022 Cochrane review found no conclusive evidence to support or reject its use and two recent trials have given mixed results.⁴³ The first, a large double-blind placebo-controlled trial of D-mannose for women with rUTI in UK

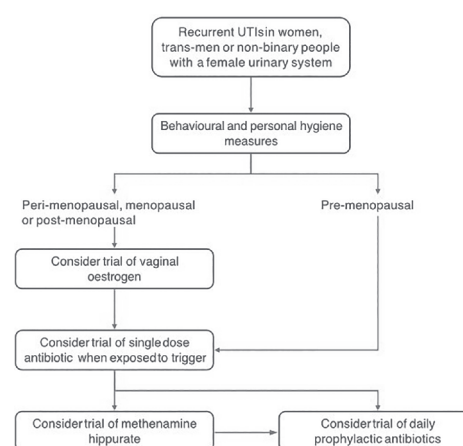


Figure 1 Flow chart of the management of rUTI adapted from the NICE Urinary tract infection (recurrent): antimicrobial prescribing guideline.¹² NICE, National Institute for Health and Care Excellence; rUTI, recurrent urinary tract infections.

primary care (n=598) found no evidence of a benefit (relative risk (RR) 0.92, 95% CI 0.80 to 1.05; unadjusted risk difference -5%, 95% CI -13% to 3%).⁴⁴ The second unblinded trial (n=75) recruiting from secondary care in Italy found for their primary outcome that the mean number of UTIs at 12 months was 0.32 UTIs/year for D-mannose, compared with 0.20 UTIs/year for prophylactic antibiotics and 1.08 UTIs/year for increasing fluid hydration.⁴⁵ They also found that time to first UTIs was longer in those randomised to D-mannose (mean=2.5 months) compared with increased hydration (1 month) but less than prophylactic antibiotics (4.5 months).⁴⁵ The 2023 Cochrane review, including 11 trials in women with rUTIs, suggested that cranberry-based products are effective in women with rUTIs (RR 0.74, 95% CI 0.55 to 0.99).⁴⁶ However, it remains unclear which preparations or dosages are optimal.

Vaginal oestrogen

Vaginal oestrogen is recommended for perimenopausal and postmenopausal women with rUTIs if self-care has not been effective.¹² Although the limited evidence for vaginal oestrogen suggests it is effective, it is off label for rUTI prevention, and there is no specific preparation recommended.¹² Therefore, the preparation should be informed by patient preferences as part of a shared decision-making discussion including the additional benefits for the genitourinary symptoms of the menopause. Risks should also be discussed; however, systemic absorption is minimal and serious side effects are rare.¹²

Research evidence, although limited, consistently demonstrates that vaginal oestrogen is effective at reducing UTI recurrence in postmenopausal women with rUTIs.^{47 48} Both a 2008 Cochrane review (n=201 patients, two trials) and a more recent systematic review (n=1936 patients, five

trials) comparing vaginal oestrogen (delivered via cream, suppositories or rings in five studies) to placebo or usual care found similar results that vaginal oestrogen is effective at reducing UTI recurrence.^{47 48} The Cochrane review included two trials, comparing vaginal oestrogen (cream or ring) to placebo or usual care, but did not pool the trial effect measures in a meta-analysis due to significant heterogeneity.⁴⁷ The more recent review did pool the trial estimates in a meta-analysis alongside three additional trials demonstrating a pooled RR of 0.42 (95% CI 0.30 to 0.59) with no evidence of increased risk of adverse effects (RR=3.06, 95% CI 0.79 to 11.90).⁴⁸ The Grading of Recommendations Assessment, Development and Evaluation score for the evidence was moderate. Depending on the preparation, this equates to a number needed to treat (NNT) for vaginal oestrogen of between 2 and 4.¹² Systemic oestrogen has not been shown to be effective at preventing UTI recurrence in women with rUTIs in both reviews based on three trials (n=2875) and is not recommended for rUTI prevention.^{47 48}

Methenamine hippurate

Methenamine hippurate, a urine antiseptic, is recommended as an alternative to daily antibiotic prophylaxis, if behavioural methods, trigger-related antibiotics (see below) and/or vaginal oestrogen (if appropriate) have all been ineffective.¹² In the urinary tract and in an acidic environment, methenamine is converted to formaldehyde and is bactericidal.³³ Based on this mechanism of action, over-the-counter urine-alkalising agents such as potassium citrate are not recommended as they may impact methenamine’s effectiveness.¹² The guidance does not discuss whether urine acidification (eg, co-prescribing methenamine with ascorbic acid (vitamin C)) is required.¹² However, the joint guideline from the American Urological

Table 3 Prophylaxis antibiotic choices, dosages and additional considerations¹²

Antibiotic	Dosage	Additional considerations
First-choice oral antibiotics for prophylaxis		
Trimethoprim	Single-dose prophylaxis 200 mg single dose after a UTI trigger (eg, sexual activity), Daily prophylaxis 100 mg once daily at night	Teratogenic risk in the first trimester of pregnancy. High levels of resistance in the UK (approximately 30%). ³³⁹
Nitrofurantoin	Single-dose prophylaxis 100 mg single dose after a UTI trigger Daily prophylaxis 50 or 100 mg once daily at night	Requires an eGFR ≥45 mL/minute for prophylaxis. Risk of hepatic and lung adverse drug reactions ⁶⁶ (eg, fibrosis). Patients should be counselled on these risks, including the signs and symptoms suggestive of respiratory and liver dysfunction and the need for liver function monitoring. Off-label for prevention of recurrent upper UTI or complicated lower UTI as nitrofurantoin does not achieve therapeutic levels in renal tissue. ⁶⁷
Second-choice antibiotics		
Amoxicillin	Single-dose prophylaxis 500 mg single dose after a UTI trigger Daily prophylaxis 250 once daily at night	High levels of resistance in the UK (between 47% and 51%). ³³⁹ Off-label for rUTI prevention.
Cefalexin	Single-dose prophylaxis 500 mg single dose after a UTI trigger Daily prophylaxis 125 once daily at night	Off-label for rUTI prevention.

eGFR, estimated glomerular filtration rate; rUTI, recurrent UTI; UTIs, urinary tract infections.

Association (AUA), Canadian Urological Association (CUA) and Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (SUFU) advises against co-prescribing ascorbic acid with methenamine since it does not significantly reduce urine pH or improve the effectiveness of methenamine.^{11 12}

Methenamine use for rUTI prophylaxis should be reviewed within 6 months of initiation and then on an annual basis.¹²

Evidence for methenamine is also limited and the NICE update focused on evidence from two recent trials based in secondary care comparing methenamine to prophylactic

Table 4 Comparison of NICE guidance to other international guidelines

		NICE guidelines	EAU guidelines	AUA, CUA and SUFU guidelines
Diagnosis of rUTIs	Urine investigations	Suspected UTIs (symptom-based±urine dipstick) with urine culture for antibiotic susceptibility testing but not explicitly required for rUTI diagnosis	Urine culture required for rUTI diagnosis	Requires typical UTI symptoms, evidence of inflammation (ie, pyuria) and uropathogen growth on urine culture.
	Other investigations	Advises considering further investigation, such as an ultrasound, where self-care and non-antibiotic forms of prevention have failed, and prior to initiation of prophylactic antibiotics. Also advises referral/specialist input if an underlying cause is suspected or the cause is unknown.	Advises against routine investigation (ie, imaging and cystoscopy) in women under 40 years without risk factors unless an underlying cause is suspected	Similar to EAU in women with rUTIs without risk factors unless poor response to treatment or an underlying cause is suspected.
Management	Behavioural	Advises enough fluid intake to avoid dehydration and behavioural and personal hygiene measures	Advises increased fluid intake in pre-menopausal women drinking less than 1.5 L States limited evidence for behavioural/personal hygiene measures but still advises to discuss them	Advises increased fluid intake in those drinking less than 1.5 L but does not include behavioural/personal hygiene measures as they state they do not have a role in rUTI prevention and 'reframing' discussions around hygiene practice has potential benefits to patients and healthcare.
	D-mannose	Advises patients 'may wish to try' but does not explicitly recommend	Recommend to use but to inform patients of the weak and contradictory effectiveness evidence	Recommends that patients should be informed that it might not be effective on its own
	Cranberry-based products	Advises patients 'may wish to try' but does not explicitly recommend	Recommends to advise on its use but to inform patients of weak and contradictory evidence.	Recommends offering as an option.
	Vaginal oestrogen	Recommended for perimenopausal, menopausal and postmenopausal women with rUTIs	Recommended for postmenopausal women	As per NICE guidelines
	Methenamine hippurate	Recommend its use but advise specialist input in those with recurrent upper or complicated lower UTIs	Recommend its use in women without urinary tract abnormalities	Recommends that it may be offered.
	Antibiotic prophylaxis	Recommends use when non-antibiotic prevention has failed (single dose related to trigger prior to daily if trigger identified)	Recommends use when non-antibiotic prevention has failed	Recommends use after discussion of benefits, risks and alternatives

AUA, American Urological Association; CUA, Canadian Urological Association; EAU, European Association of Urology; NICE, National Institute for Health and Care Excellence; rUTI, recurrent UTI; SUFU, Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction; UTIs, urinary tract infections.

Box 3 Information resources

For women with recurrent urinary tract infections (UTIs) Bladder Health UK <https://bladderhealthuk.org/>

UK charity offering support for people with bladder related problems, including recurrent UTIs

TARGET UTI patient information leaflets <https://elearning.rcgp.org.uk/mod/book/view.php?id=12652> Patient information leaflets in various formats on UTIs including information on when to seek urgent help. Leaflets are also translated into various different languages too (<https://elearning.rcgp.org.uk/mod/book/view.php?id=12647&chapterid=443>)

For healthcare professionals

UKHSA and NHS England UTI Diagnostic guide <https://www.gov.uk/government/publications/urinarytract-infection-diagnosis> Quick reference guides for diagnosing UTIs in women under and over 65 years and in those using catheters.

SIGN Management of suspected bacterial lower urinary tract infection in adult women https://www.sign.ac.uk/media/1766/sign-160-uti-0-1_web-version.pdf Also provide guidance on diagnosing UTIs in women under and over 65 years and in those using catheters.

TARGET "How to...?" guide for recurrent UTIs

<https://elearning.rcgp.org.uk/mod/book/view.php?id=12649>

Resources to support healthcare professionals in primary care review patients with recurrent UTIs. for reviewing patients on prophylactic antibiotics for recurrent UTIs

NHS, National Health Service; SIGN, The Scottish Intercollegiate Guidelines Network; TARGET, Treat Antibiotics Responsibly, Guidance, Education and Tools; UKHSA, UK Health Security Agency.

antibiotics. The first was a non-inferiority trial (n=240) with a mean age of 50 years and a mean number of seven UTIs prior to enrolment.³³ They found at 12 months that methenamine was non-inferior (based on an inferiority margin of one UTI/year informed by those with lived experience of rUTI), with an absolute difference in UTI incidence rate of 0.49 (95% CI 0.08 to 0.90).³³ The UTI rates increased in both arms in the 6 months following cessation of prophylaxis but remained lower than pre-prophylaxis UTI rates, with over 50% of all trial participants remaining UTI free.⁴⁹ The second trial was US-based (n=92), with a mean age of 72 years and on average four UTIs/year prior to enrolment.⁵⁰ They found at 12 months, no difference in the number of women with a further UTI recurrence between the arms (65% in both arms).⁵⁰ Subsequently, a trial focussing on older women (70 years and older) with rUTIs has been published.⁵¹ It was conducted in several European countries and compared methenamine to placebo with an average of 3.5 UTIs in the 12 months prior to study entry.⁵¹ They found methenamine resulted in fewer antibiotic-treated UTIs at 6 months compared with placebo (incidence rate ratio (IRR)=0.75, 95% CI 0.57 to 1.00). However, during the 6 months of follow-up, after the methenamine and placebo were stopped, the UTI rate increased in both arms, but was greater in the methenamine arm (IRR=1.7, 95% CI 1.3 to 2.3).⁵¹

Antibiotics

NICE CKS suggests that in women who can self-diagnose and treat their UTIs, a self-initiated acute, short course of antibiotics can be used.²² NICE guidance on antibiotic prophylaxis includes two forms. These include a single dose of antibiotics taken after a UTI trigger (eg, sexual activity), or daily

Table 5 Information for patients—understanding your options for preventing recurrent UTIs

Option	Benefits (Why consider it?)	Risks (What could go wrong?)
Self care and lifestyle changes (eg, drinking more fluids, hygiene advice)	▶ Easy to try at home ▶ May reduce UTIs, especially with increased fluid intake	▶ May not be enough on its own ▶ Limited evidence for hygiene advice ▶ Focus on adequate hydration and avoid overhydration.
Vaginal oestrogen (for perimenopausal and postmenopausal women)	▶ Effective at reducing the number of UTIs ▶ May also help with dryness or discomfort that occurs due to the menopause	▶ Mild irritation or spotting ▶ Not officially licensed for UTI prevention ▶ Little is absorbed into the bloodstream so the risks are low.
Methenamine hippurate (a non-antibiotic tablet)	▶ Effective at preventing UTIs ▶ Not an antibiotic so reduced risk of antibiotic resistance	▶ Can cause stomach upset ▶ Doesn't work for everyone ▶ Not suitable for people with a history of gout or may not be suitable if kidney problems
Antibiotic prophylaxis (taken daily or after a known trigger)	▶ Effective way to prevent UTIs ▶ Can be tailored to your situation, for example, if UTIs are related to sexual activity	▶ Side effects (eg, nausea, allergic reactions). Some side effects can be serious and life-threatening due to lung and liver scarring. ▶ Increases the risk of antibiotic resistance.

UTIs, urinary tract infections.

prophylactic antibiotics for rUTI prevention (table 3). Single-dose antibiotic prophylaxis is advised if there is an identified trigger and if behavioural measures and vaginal oestrogen are ineffective or not appropriate.¹² Daily antibiotic prophylaxis should be considered if the above non-antibiotic methods of prevention have failed or are not appropriate. The individual's preferences for antibiotics should be considered in both cases as well as whether further investigation is needed to identify a cause for the rUTIs (eg, an ultrasound scan).¹² Key aspects for consideration and discussion prior to initiation include:

- how to use the antibiotic (ie, for single or daily dose prophylaxis),
- the severity and frequency of UTIs,
- the risks associated with long-term antibiotic prophylaxis including adverse drug reactions (especially the serious adverse drug reactions associated with nitrofurantoin use (see table 3) and the risk of developing antibiotic-resistant infections,
- previous urine culture results and antibiotic susceptibility,
- recent antibiotic use as this might influence antibiotic resistance,
- the need for a review of antibiotic prophylaxis every 6 months,
- the individual's preferences around the use of antibiotics.

A 2004 Cochrane review (n=430 patients, ten trials) demonstrated that prophylactic antibiotics are effective at reducing UTI recurrence in women with rUTIs, compared with placebo, although the quality of studies was reported as 'poor'.⁵² The review demonstrated that prophylactic antibiotics, compared with placebo, reduced the proportion of women experiencing at least one UTI recurrence during follow-up based on either clinical (RR 0.21, 95% CI 0.13 to 0.33) or microbiological criteria (RR 0.15, 95% CI 0.08 to 0.28). This equates to an NNT of 2.20 (95% CI 1.80 to 2.80) based on clinical criteria and 1.85 (95% CI 1.60 to 2.20) based on microbiological criteria.⁵² However, concerns persist regarding adverse effects and the potential for uropathogens to develop resistance, which may complicate future treatment as demonstrated in several trials and observational studies.^{33 52–55} In a recently published observational study, it was estimated that prophylactic antibiotics increased the absolute risk of developing an antibiotic-resistant uropathogen on urine culture by 6.9% to one antibiotic and to two or more antibiotics by 8.0%.⁵⁶ This equates to a number needed to harm of 14.6 and 12.5, respectively.⁵⁶

Antibiotic prophylaxis should be reviewed every 6 months, including an assessment of its effectiveness, whether to continue, stop or change the form of prevention (factoring in the individual's preferences and the risk of antibiotic resistance) and re-discussion about self-care.¹² If prophylaxis is stopped, then patients should have rapid access to acute treatment for a further UTI.

Comparison to other guidelines

There are several published international guidelines on the management of rUTIs, which differ from the NICE guidelines just discussed. These include the European Association of Urology (EAU) guidelines on urological infections and the recently updated joint guideline of recurrent uncomplicated UTIs in women from the AUA, CUA and SUFU.^{10 11} How these guidelines compare and the key differences are highlighted in table 4.

Uncertainties and emerging research

There remain several uncertainties in rUTIs and their management. As discussed, diagnosis is challenging due to the limitations of current diagnostics and there is a need for more accurate and

rapid tests. For example, in women over 65 years, urine dipstick is unhelpful and urine culture results are not available at the point of assessment. More rapid tests would allow test-informed decision making and prescribing. This could reduce unnecessary prescribing, but more importantly, would help ensure the causative uropathogen is susceptible to the antibiotic prescribed, thereby reducing the risk of treatment failure. Point-of-care tests for bacteriuria and antibiotic susceptibility are under development, and there is ongoing research to assess their diagnostic accuracy and clinical effectiveness.^{57 58} The role of the urinary and vaginal microbiome is another area of uncertainty, and their role in the pathogenesis of UTIs and rUTIs. Linked to this is the potential role of probiotics in rUTI prevention. Existing evidence demonstrates limited benefit; however, few studies have rigorously assessed the clinical and cost effectiveness of probiotics.⁵⁹ There are also potentially new and emerging treatment options that show promise. Early research suggests that UTI vaccines could have potential benefits in the short term, but it remains unclear if they have long-term benefits.^{10 60} Finally, bacteriophage therapy has also emerged as a promising option, whereby viruses are deployed to target and destroy specific bacteria, although evidence is limited in the context of rUTIs.^{10 61}

Conclusions

Recurrent UTIs affect a substantial proportion of women and contribute to significant morbidity and frequent engagement with healthcare. This article has summarised the challenges related to diagnosis and management, highlighted the updated NICE guidance and how it compares to other international guidelines, and the remaining uncertainties and the emerging research in this field. It also provides information resources for both those affected by recurrent UTIs and healthcare professionals (box 3), alongside patient-focused information on the available options to prevent recurrent UTIs (table 5). Clinicians are encouraged to re-evaluate current rUTI management strategies in light of updated NICE guidance, with particular emphasis on optimising non-antibiotic prevention, enhancing patient-centred care and strengthening antimicrobial stewardship.

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